

Zero-shot monitoring of construction workers' personal protective equipment based on image captioning

Daeyoung Gil, Ghang Lee^{*}

Department of Architecture and Architectural Engineering, Yonsei University, Seoul, South Korea

ARTICLE INFO

Keywords:

Zero-shot detection
Computer vision
Site monitoring
Image captioning
Human-pose estimation

ABSTRACT

Previous studies have deployed object-detection-based approaches to automate personal protective equipment (PPE) safety monitoring. However, previously proposed methods require large amounts of labeled data for different PPE items. This study proposes a zero-shot PPE monitoring method that does not require a training process to overcome this problem. The proposed method comprises three steps. First, it detects workers onsite from images and crops body parts using human-body key points. Next, the cropped body images are described in text using image captioning. Finally, the extracted text is compared with prompts describing body parts wearing PPE, and safety is determined based on cosine similarity. Compared to the F1-score of 73.5% achieved by traditional object detection approaches trained on 50 images for hardhat monitoring, the proposed zero-shot approach demonstrates significant improvement with an F1-score of 82.6%. It also surpasses the previous zero-shot monitoring performance (an accuracy of 53%).

1. Introduction

Personal protective equipment (PPE) is fundamental to the safety of construction workers. Naturally, several studies have been conducted to monitor the PPE compliance of onsite workers via PPE detection based on convolutional neural networks (CNN) to reduce accidents [1–4]. However, to utilize a CNN model, a large volume of data must be prepared using a time-consuming and laborious process [5–8]. Transfer learning [5,9,10] and data augmentation [11–13] have been proposed to overcome this limitation. Transfer learning significantly reduces the amount of data required to train a model for a specific task by reusing the weights of a large pre-trained model [5]. And, data augmentation alleviates the burden of constructing an image dataset by supplying additional data. Data augmentation can be achieved in two ways: by transforming existing images to increase the number of samples or by using other tools to generate entirely new data. The former involves simple geometric transformations of image data [13], whereas the latter involves generating new images from building information modeling (BIM) models [12] or generative adversarial networks (GANs) [11]. However, transfer learning and data augmentation are limited because they require significant amounts of training data for different types of PPE items, such as hardhats, vests, and gloves, to be effective.

Recently, few-shot and zero-shot learning approaches have been

adopted for site monitoring, enabling neural networks to be trained with only a few or no images [14]. Because the few-shot learning approach trains models with fewer than five to ten cases per class, the performance of few-shot object detection is lower than those of large-data-based supervised learning approaches. By comparison, zero-shot learning (ZSL), which performs a new task without training data, naturally performs less than other methods. Thus, to improve the performance of ZSL, ZSL is often combined with other methods, such as knowledge graphs (KG) [15] and prompt engineering [16]. Those approaches are studies that have well-adopted ZSL in construction, but some research applying ZSL to construction beyond these is still limited. Considering that ZSL reduces the model's data dependency and enables learning from various domains, there is a need to develop models that can easily be trained for construction sites.

In this regard, this study proposes a three-step-ZSL-based PPE monitoring (hereafter referred to as zero-shot monitoring) method, which combines pose estimation, image captioning, and semantic text similarity analysis. The proposed method receives three inputs as prompts: a construction site photo, a pair of PPE conformance statements, and a violation statement. First, the workers are extracted from the input image. From the extracted workers, body parts, such as the head, face, torso, and hands, wearing safety equipment are extracted using the pose-estimation model. Second, descriptions of each body part

^{*} Corresponding author.

E-mail address: glee@yonsei.ac.kr (G. Lee).

<https://doi.org/10.1016/j.autcon.2024.105470>

Received 2 June 2023; Received in revised form 4 October 2023; Accepted 10 May 2024

Available online 17 May 2024

0926-5805/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

are generated using an image captioning model. Finally, the similarity between the generated descriptions of the body parts and the PPE requirement text for each body part is assessed to determine whether the PPE requirements are fulfilled.

Regarding the scope of study, whereas previous studies focused mostly on hardhats [2–4,8], this research expands the scope of automated PPE compliance checking from hardhats to safety goggles, vests, and gloves to demonstrate the generalizability of the proposed method without additional training.

This paper comprises five sections. Following the introduction, related studies are reviewed to provide the research background, and the limitations of previous studies and the differentiation of this study are explained. The proposed zero-shot monitoring method is then introduced in the third section, and the verification experiments performed on various PPE items to evaluate the performance of the proposed method are recounted in the fourth section. Finally, the fifth section discusses the results, their implications, and the conclusion of this study, including its contributions, limitations, and future research directions.

2. Research background

2.1. Onsite PPE detection using computer vision

Many safety accidents at construction sites can be prevented by workers wearing proper PPE [1,17]. Consequently, several studies have been conducted to devise methods for monitoring proper PPE use by workers, based on the application of computer vision techniques to CCTV images [18,19]. Hardhat detection, an example of PPE detection, has been particularly researched, and rule-based computer vision approaches using color channels and edge features were initially utilized [20]. However, the rule-based approach has limitations in flexibly handling external factors, such as shadows or shooting angles of images. On the other hand, several studies have focused on directly visible features for the robust performance of target detection, and many have been conducted with feature-based approaches, such as scale-invariant feature transform (SIFT) and histogram of gradient (HoG). For example, Rubaiyat et al. [21] proposed an advanced method that matches the color channel to the image features from the HoG descriptor. HoG has the strength of feature matching for moving objects and has also been applied to other research studies on safety monitoring by tracking hardhats and human bodies [18,19]. However, it has been difficult to obtain robust performance because it is challenging to recognize objects using these methods semantically [22]. Thus, since the introduction of CNN, many studies have adopted these networks for monitoring solutions.

Recent studies that have utilized CNN for object detection have largely been based on Faster Region-based CNNs (R-CNNs) [23], and You Only Look Once (YOLO) [24]. They enable the recognition of target objects, usually hardhats or workers, on construction sites and exhibit robust and improved performance when trained with sufficiently large datasets. For example, Fang et al. [2] suggested a hardhat detection module based on Faster R-CNN that demonstrated robust performance regardless of photo range, time, and weather changes. Alternatively, Shen et al. [4] recently proposed an advanced hardhat detection network based on the DenseNet architecture [25] and demonstrated high-end performance.

To train a CNN, a large labeled dataset and, correspondingly, massive calculation effort and time are required. As outlined in Table 1, the previous CNN methods developed for the construction field required at least 400 images per class. Quite notably, Shen et al. [4] utilized 9800 images to create their dataset, whereas Fang et al. prepared over 100,000 images to train their model [2]. Methods for generating photorealistic images from BIM models [26] or extracting frames from videos [4] have been deployed to supplement these image datasets. Both methods enhance performance; however, supervised learning requires large and diverse datasets to improve performance. Transfer learning

Table 1

Data sizes of CNN-based field monitoring research in the construction field.

#	Author	Target object	Number of classes	Number of images per class	Image source
1	Kim et al. [5]	Construction equipment	5	353–787	ImageNet
2	Nath et al. [9]	Workers, buildings, and equipment	3	517–715	Web mining
			7	21–633	Web mining
3	Kolar et al. [26]	Safety guardrail	2	2000	Synthetic images
4	Shen et al. [4]	Safety helmet	2	18,800	Construction videos
5	Zheng et al. [27]	Modules of modular construction	2	100–1000	Synthetic images
6	Bunrit et al. [28]	Construction materials	3	400	(Unspecified)
7	Zhu et al. [29]	Construction defects	5	48–510	Construction site photos
8	Hayat et al. [3]	Safety helmet	2	5000	(Unspecified)

[5,9,41] can reduce the burden of constructing a large dataset but still requires hundreds of images and additional training to adapt the model to new domains or tasks.

As standard datasets for general objects, such as ImageNet [30] and MS COCO [31], available datasets for specific objects, such as hardhats and construction equipment, have also been constructed in previous studies [5–8,32]. However, such datasets are thus far limited to hardhats only; to the best of the authors' knowledge, similar data for similar purposes are yet to be made available for other PPE, implying that a customized dataset is essential for effectively applying PPE safety monitoring. However, given the challenge of gathering and labeling images for such a dataset, it is necessary to reduce their dependence on the dataset [33]. Furthermore, various pre-trained models have recently emerged, and techniques for applying those weights to new domains have already been proposed. Adapting these pre-trained models to the construction field could reduce dependence on specific datasets and ensure flexibility in the model configuration.

2.2. Few-shot and zero-shot learning

As mentioned earlier, attempts have been made to eliminate the need for training data and enhance the domain adaptability of artificial intelligence models. In few-shot learning, the model learns the meta-features of novel class images from a few sample images. Few-shot learning methods include distance-based learning, such as Siamese networks [34], proto-typical networks [35], and optimization-based learning methods, such as model-agnostic meta-learning [36]. In the construction field, Kim and Chi [14] suggested a few-shot detection module for construction equipment using metadata of features extracted from the AIM dataset [5]. Cui et al. [37] also proposed a few-shot detection method. They demonstrated a way of optimizing the similarity between defect images by applying cross and contrastive loss to the objective function.

Unlike few-shot learning, which requires a small number of samples (shots), ZSL makes assumptions regarding unseen classes without any given data [38]. In ZSL, information about the unseen class is provided in ways other than images, such as through the relationship between pre-trained seen classes [39], among other various ways. For example, Kampffmeyer et al. [15] utilized a knowledge graph. By updating the relationships between classes during training, the model can infer the label of an input image using the meta-information of the seen classes.

Although it requires graph updates, this approach enables the model to learn by providing only the relationships necessary for inferring new labels without requiring training images. This strategy has also been utilized in the construction domain, where Pan et al. [40] adopted a knowledge graph to extract unseen entities. Another ZSL approach uses a large multimodal pre-trained model [41].

Contrastive language–image pre-training (CLIP) [16] is a representative model of vision language pre-training (VLP), which enables the model to train an unseen class using only its description in the text. CLIP, introduced by Radford et al. [16], can learn the information of an unseen class using textual inputs. This model consists of two encoders for text and image inputs, which transfer the relationship between the text and image features and can infer unseen image classes based on the observed image features. However, VLP models pre-trained with general objects perform poorly on domain-specific photos, such as hardhats; therefore, an additional process is necessary for site applications. It is necessary to optimize for more appropriate text through prompt engineering to utilize VLP models such as CLIP in the construction domain [42]. In construction, Yong et al. [43] employed prompt engineering of CLIP for building defect detection and provided general guidelines for making prompts. However, our preliminary study indicates that prompt-engineering-based ZSL yielded poor results for PPE monitoring, suggesting the need for a new approach.

2.3. Research gaps

Recent studies on construction safety monitoring have been based on traditional CNN-based-object-detection methods [3,4]. They showed improved performance compared to the rule-based approach, but they required the preparation of large datasets, which would demand considerable human labor [5–8,32]. Although few-shot and ZSL have been proposed to reduce the dependence on large datasets, only a few studies [14,37] have been conducted on these approaches, and their application in the building construction domain is in its nascent stage. In particular, applying these techniques for onsite safety monitoring has not yet been attempted.

Although several studies in the construction domain have attempted to reduce the burden of creating datasets by utilizing metadata for few-shot or ZSL, further processing is still required. For example, the ZSL approach based on a predefined knowledge graph [40] or ontology will require relation updates when new classes are added, which can lead to an additional workload. The KG-based ZSL relies on the relationships of the base classes used in pre-training, which can burden model updates as new labels are added when the model becomes larger. On the other hand, text-based ZSL approaches [16] have the advantage of obtaining information on unseen classes and can provide semantic information about words through prompts alone. However, they cannot deal with domain-specific knowledge directly because they are pre-trained on general objects and unfamiliar with technical terms. For specific items such as construction PPE, additional tasks are required to induce the intended information to be learned by the model.

3. Proposed method

In the initial attempt to detect PPE use by workers, CLIP was fully applied to the entire process with only text prompts because this method was designed for text-based ZSL. However, the proposed approach did not successfully produce acceptable performance in detecting detailed objects, such as hardhats, via simple label transfer. Therefore, a target-specific approach was attempted while maintaining a large framework in a similar manner. After the caption text was extracted from the target-specific-body-part images, text embedding via a language model was applied to determine whether the PPE was worn. Therefore, the proposed zero-shot PPE monitoring system primarily comprises three steps: 1) body-part extraction using pose estimation, 2) image captioning, and 3) text similarity comparison, including a step to generate text for

compliance and violation cases from the Occupational Safety and Health Administration (OSHA) PPE regulations [44]. Fig. 1 illustrates these three steps. The entire process utilized pre-trained weights for general objects, and direct training of the model was not conducted except for the prompt engineering process in the green-colored area. The following subsections describe these three steps in detail.

3.1. Body-part extraction

Checking PPE use on workers begins with identifying workers in a site image and extracting each worker's skeletal posture key points. Fig. 2 illustrates the body-part extraction process. Construction workers in an input photo were detected using YOLO v5 for object detection in the human category [24], and human skeletal key points were extracted using a pose-estimation model, mm-Pose [45], based on HRNet [46]. The pose estimation was pre-trained using the 2017 COCO dataset, which consists of 17 body key points. Using the body key points, the body parts are divided into four parts—namely, the head, face, torso, and hand—to detect hardhats, goggles, safety vests, and gloves to achieve the aim of this study. Nine body key points for the nose, shoulders, hips, wrists, and elbows are used to define the bounding box for each body part. For example, the bounding box of a head is defined as a square with a nose at the center and a width equal to the distance between the two shoulders. An additional buffer space of 30% of the distance was added to the head image to ensure that the entire hardhat was included, thereby decreasing the ratio for the face image. Other body parts were extracted using the same process. The reference key points and the added buffer spaces for each body part are listed in Table 2.

3.2. Image captioning

A description of the cropped body-part image was obtained using ClipCap [47]. An image captioning model based on CLIP image captioning is applied to a construction photo on three levels to demonstrate this: the entire construction photo, a full-body shot, and a body-part shot (Fig. 3). Our intention with image captioning was to extract words related to PPE and increase their relevance to prompts. As illustrated in Fig. 3, the automatically generated caption at the body-part level was closest to the intended caption. Therefore, body-part extraction was performed as the first step.

The pose-estimation model [45] uses pre-trained weights from the MS COCO dataset [31], and ClipCap [47] uses the Conceptual Captions dataset [48].

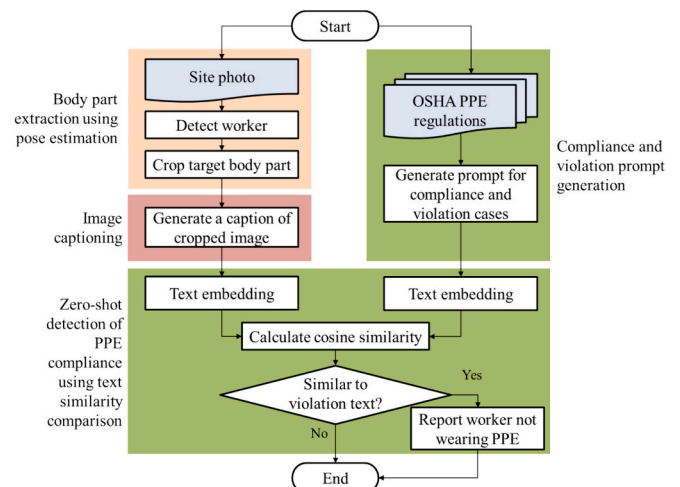


Fig. 1. Overall flow of proposed zero-shot PPE monitoring method.

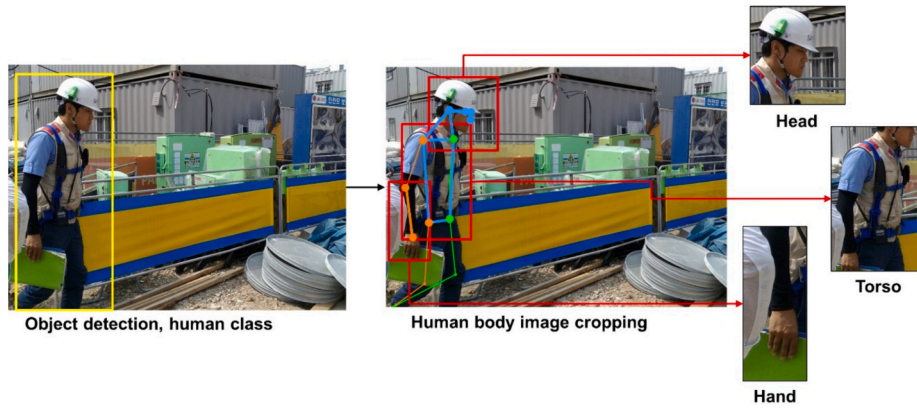


Fig. 2. Illustrated example of the body-part extraction process.

Table 2

Reference key points for defining the bounding box of a body part.

Body part	Center point	Length of target boundary box	Added buffer space
Head	Nose point	Distance between shoulders	30%
Face	Nose point	Farthest distance between shoulders and hips	10%
Torso	Center point of both shoulders and hips	Distance between elbows and wrists	15%
Hands	Center points of elbows and wrists		

3.3. Detection of PPE compliance among workers

In this study, to determine whether a worker complies with OSHA PPE regulations [44], the similarity between the automatically generated image caption and prompts prepared based on OSHA PPE regulations was measured using cosine similarity. The OSHA defines PPE regulations as follows:

- 1910.135(a)(1): The employer shall ensure that each affected employee wears a protective helmet when working in areas where there is a potential for injury to the head from falling objects.
- 1910.135(a)(2): The employer shall ensure that a protective helmet designed to reduce electrical shock hazard is worn by each such affected employee when near exposed electrical conductors which could contact the head.
- 1926.100(a): Employees working in areas where there is a possible danger of head injury from impact, or from falling or flying objects,

or from electrical shock and burns, shall be protected by protective helmets.

- 1926.100(b)(2): The employer must ensure that the head protection provided for each employee exposed to high-voltage electric shock and burns also meets the specifications contained in Section 9.7 (“Electrical Insulation”) of any of the consensus standards identified in paragraph (b)(1) of this section.

The results of zero-shot detection vary significantly depending on the composition of prompts [49–52]. The process of optimizing prompts is referred to as prompt engineering. Radford et al. [16] proposed a standard form for constructing prompts: “a <category> photo of <label>.” Building on this, Yong et al. [43] discovered that prompts composed of a complete sentence rather than keywords and domain-specific definitions rather than general ones perform better in domain-specific zero- and few-shot classifications. Building upon the prompt engineering methods for text-based ZSL demonstrated in previous studies. This study aims to tailor domain-specific information to the output structure.

For the similarity comparison, we initially considered detecting a worker’s PPE compliance using a single prompt, either a compliance or violation prompt. However, if only one prompt is compared with an automatically generated image caption, a threshold value must be set to determine the worker’s PPE compliance. Therefore, the proposed method compares automatically generated image captions against a compliance prompt and a violation prompt as a pair. Suppose an automatically generated image caption is more similar to a compliance prompt than a violation prompt; the worker is deemed to be wearing a hardhat, and vice versa. This approach prevented us from determining a threshold value randomly or based on many experiments.

To fulfill the purpose of this study and to improve the accuracy of our

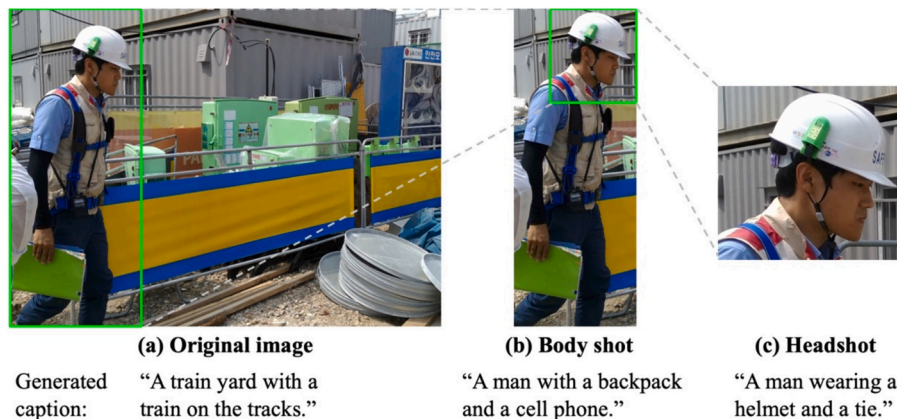


Fig. 3. Examples of varied captioning outputs for different levels of focus.

PPE compliance checking method, the PPE regulation was first simplified to “A person shall be protected by protective helmets.” This simplification was performed by removing the reasons for wearing protective helmets and generalizing the subject from “employee” to “person” because an image captioning model would not recognize a contractual relationship [44].

ClipCap, the image captioning model used in this study, outputs the sentences in the form “a <Subject> <Predicate> <Object>.” We attempted to compose prompts that properly reflected the OSHA PPE regulations. First, we used “a person” as the subject to apply gender neutrality. For the predicate [44], “with” and “wearing” were used to indicate whether a PPE item was worn. For the <Object> part, target PPE items and their synonyms from the Collins dictionary [53] were used. For example, OSHA also refers to the “head protection” equipment as a “protective helmet.” Their synonyms from Collins are “hardhat” and “protective headgear.” In addition, to compare the results with cases wherein entire sentences were included without changes in the sentence structure, one OSHA sentence was extracted and used as in the experiment. Table 3 summarizes the terms used in this study, including the code numbers and sources. These terms generated prompts aligned with OSHA PPE regulations, as described earlier.

As previously mentioned, we set prompts specifically for detecting hardhat wearers. However, since the actual goal of our study is to identify individuals not wearing PPE rather than those wearing them, a violation prompt for this purpose has been created. Two approaches were used in this study. The first was to change the aforementioned compliance prompts into their corresponding negative forms (SPO1, SP1O1-SP2O4). In this approach, violation prompts are constructed by replacing the <Predicate> parts with “without” and “not wearing,” respectively. In the second approach, compliance prompts describe people not wearing hardhats (SPO1, SP1O1-SP2O4), in contrast to compliance prompts that refer to people wearing hardhats (SPO2v, SPO3v). With these approaches, the aim is to test whether the negative sentence prompts work appropriately and, if not, to use the affirmative sentences instead. A pair of compliance-violation prompts are required to conduct the test, but there are no corresponding compliance prompts for SPOv2 and SPOv3, representing not wearing safety helmets in affirmative sentences. Thus, we conducted experiments using SPO1c as a substitute pair. We generated eleven prompt pairs by combining the subject, predicate, and object terms identified earlier. Table 4 presents the prompts generated by combining the terms in Table 3. The violation prompts were constructed by replacing the predicates “with” and “wearing” with “without” and “not wearing,” respectively.

The process proposed in this study can be summarized as illustrated in Fig. 4. It takes a prompt consisting of a pair of compliance and violation and an onsite image as input and finally outputs the PPE monitoring results of workers. The image data are converted into text through pose estimation and image captioning and are then embedded for comparison with the input prompt using cosine similarity after text embedding. If the cosine similarity between the captioning output and the violation prompt is higher than that between the captioning output and the compliance prompt, the worker’s PPE is determined to be not worn.

For text similarity comparison, this study used sentence bidirectional

Table 3
Terms used to generate prompts and sources.

Subject (S)	Predicate (P)	Object (O)	Source
SPO1: A person shall be protected by protective helmets			OSHA
S: A person	P1: with	O1: a protective helmet	OSHA
	P2: wearing	O2: a hardhat	Collins
		O3: protective headgear	Collins
		O4: a hardhat, protective gear, or a helmet	OSHA and Collins

Table 4
Prompts to determine PPE compliance among workers.

	Code	Compliance prompt	Code	Violation prompt
OSHA vs. violation statements	SPO1c	a person shall be protected by a protective helmet.	SPO1v	a person is not protected by a protective helmet.
			SPO2v	a person with hair and no head protection
			SPO3v	a person without a protective helmet
	SP1O1c	a person with a protective helmet	SP1O1v	a person without a protective helmet
With vs. without	SP1O2c	a person with a hardhat	SP1O2v	a person without a hardhat
	SP1O3c	a person with protective headgear	SP1O3v	a person without protective headgear
				a person without a hardhat, protective headgear, or a helmet
	SP1O4c	a person with a hardhat, protective headgear, or a helmet	SP1O4v	a person not wearing a protective helmet
	SP2O1c	a person wearing a protective helmet	SP2O1v	a person not wearing a hardhat
	SP2O2c	a person wearing a hardhat	SP2O2v	a person not wearing protective headgear
Wearing vs. not wearing	SP2O3c	a person wearing protective headgear	SP2O3v	a person not wearing a hardhat, protective headgear, or a helmet
	SP2O4c	a person wearing a hardhat, protective headgear, or a helmet	SP2O4v	

encoder representations from transformers (SBERT) [54]. SBERT, backboned on MPNet [55], embeds the descriptive texts of cropped images and prepares prompts as vectors of size (1768).

4. Experiments and analysis

4.1. Datasets for validation

For this study, the proposed method was validated using hardhat images provided by Harvard Dataverse [56]. For the experiment, 600 images were selected from the image dataset, including 300 workers wearing hardhats and an equal number of images of workers not wearing hardhats. Workers’ compliance with hardhat regulations was checked through tests that determined whether each worker in the images wore a hardhat. In the selection process, photos that included worker images smaller than 50×50 pixels were excluded because they were exceedingly small for the human eye to check for compliance. However, regarding other PPE items, including safety vests, goggles, and gloves, applicable datasets are not yet publicly available. Therefore, we manually constructed a small dataset with 100 labeled images to test whether the proposed method can be applied to other PPE items.

4.2. Prompt engineering

As listed in Table 4, eleven different initial pairs of compliance prompts and their corresponding violation prompts were prepared. However, the initial pairs of compliance and violation prompts, as shown in Table 4, were not necessarily the optimal pairs of prompts that could yield the best performance. This study adopted a different approach, requiring one run of 11 experiments to determine the optimal pairs of compliance and violation prompts, instead of using a brute-force approach to compare the performance of all 9×11 combinations of compliance and violation prompts.

As the goal of PPE monitoring is to detect violation cases, the

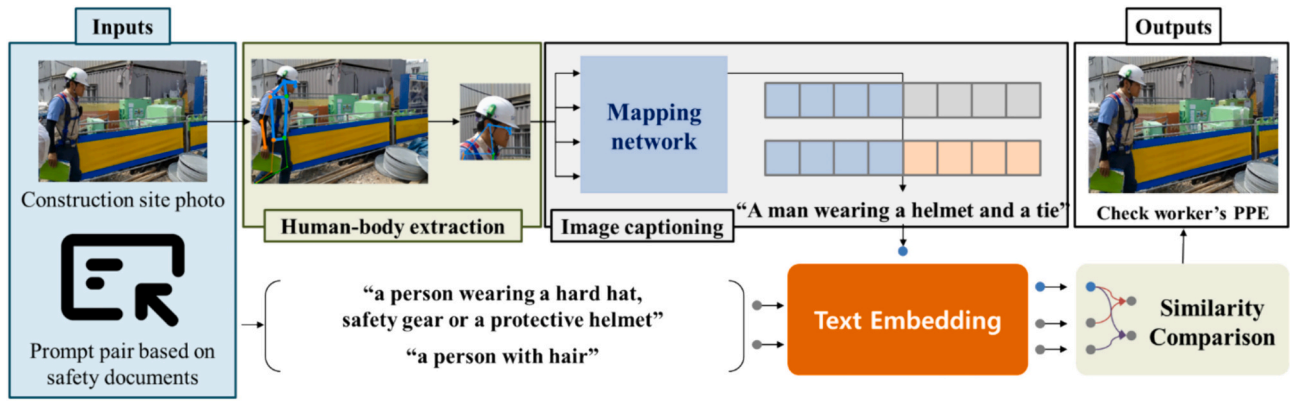


Fig. 4. Summarized process of the proposed method.

violation prompts are set to be positive and always related only to the results that are judged as positive; the correctly detected violation cases are true positive (TP), and the incorrectly detected violation cases are false positives (FP). On the other hand, the compliance prompts are related only to true negative (TN) and false negative (FN) by the same token. Thus, in terms of the performance of the violation prompt, the one with the largest difference in mean cosine similarity values (mCS) between the TP and FP cases ($mCS_{TP} - mCS_{FP}$) is the violation prompt with the highest performance. In the case of compliance prompts, the one with the largest difference in mCS between TN and FN cases ($mCS_{TN} - mCS_{FN}$) is the compliance prompt with the highest performance. Naturally, the pair comprising the best-performing violation and compliance prompts forms the best pair.

Table 5 shows the accuracy and mean cosine similarity for each prompt. Based on the scores for the various prompts, the best pair was a combination of the compliance prompt SP204c and the violation prompt SPO1v [“a person wearing a hardhat, head protection, protective gear, or a helmet,” “a person with hair”]. The pair [SP104, SP204] resulted in the largest difference between true and false positive values ($mCS_{TP} - mCS_{FP}$) and the largest difference between the true and false negative values ($mCS_{TN} - mCS_{FN}$). Separately combining them is possible because each prompt’s performance is independent.

Table 6 shows the performance of the optimized prompt pair

[SP204c, SPO2v] against the other pairs of compliance and violation prompts. The other pairs were those generated by grouping an affirmative prompt and the negated prompt of the affirmative prompt listed in Table 4.

Our study aimed to detect hardhat non-wearers; thus, we set the detection of hardhat non-wearers through violation prompts as TP and the detection of wearers as TN. Three metrics were used for the validation tests: accuracy, F1-score, and F2-score. In safety studies, the F2-score is more crucial because it places twice as much weight on recall than on precision due to the prioritization of worker safety; identifying as many hardhat non-wearing workers as possible is regarded as more important, even if some cases turn out to be false negatives (workers mistakenly identified as PPE regulation violators).

The optimized pair of prompts exhibited an accuracy of 81.50%, F1-score of 82.57%, and F2-score of 85.56%, whereas the other pairs yielded an accuracy of 60.67%, F1-score of 66.19%, and F2-score of 72.63% at the highest.

The results also revealed several guidelines for the construction of prompts.

- In the case of the compliance prompts, there was not much difference between using “with” (SP101c to SP104c) and using “wearing” (SP201c to SP204c) in the <Predicate> part of the prompt.

Table 5

Accuracy and mean cosine similarity for each prompt.

Prompt pair	Mean Cosine similarity (mCS)				$(mCS_{TN} - mCS_{FN})$	$(mCS_{TP} - mCS_{FP})$
	mCS_{TP}	mCS_{FP}	mCS_{FN}	mCS_{TN}		
[SPO1c, SPO1v]	0.1526	0.2699	0.1571	0.2932	0.1361	-0.1173
[SPO1c, SPO2v]	0.3949	0.3687	0.1571	0.2932	0.1361	0.0262
[SPO1c, SPO3v]	0.2778	0.3110	0.1571	0.2932	0.1361	-0.0332
[SP101c, SP101v]	0.1470	0.2606	0.2831	0.4309	0.1478	-0.1136
[SP102c, SP102v]	0.2176	0.3232	0.3262	0.4687	0.1425	-0.1056
[SP103c, SP103v]	0.2164	0.3116	0.3750	0.4760	0.1010	-0.0952
[SP104c, SP104v]	0.2109	0.3459	0.2966	0.4593	0.1627	-0.1350
[SP201c, SP201v]	0.1239	0.2284	0.2770	0.4285	0.1515	-0.1045
[SP202c, SP202v]	0.2250	0.3229	0.3213	0.4671	0.1458	-0.0979
[SP203c, SP203v]	0.1836	0.2758	0.3617	0.4709	0.1092	-0.0922
[SP204c, SP204v]	0.2039	0.3206	0.2959	0.4597	0.1638	-0.1167

Table 6

Performance of optimized prompt combination compared to the other pairs.

Prompt Pair		Accuracy	F1-score	F2-score
Optimized	[SP204c, SP101v]	81.50%	82.57%	85.56%
OSHA vs. violation statements	[SPO1c, SPO1v]	60.33%	66.19%	72.63%
	[SPO1c, SPO2v]	50.17%	3.24%	2.07%
	[SPO1c, SPO3v]	60.67%	35.52%	25.67%
With vs. without	[SP101c, SP101v]	49.67%	0.66%	0.42%
	[SP102c, SP102v]	50.00%	1.96%	1.24%
	[SP103c, SP103v]	49.67%	0.00%	0.00%
	[SP104c, SP104v]	50.33%	2.61%	1.66%
	[SP201c, SP201v]	49.50%	0.00%	0.00%
Wearing vs. not wearing	[SP202c, SP202v]	49.83%	1.95%	1.24%
	[SP203c, SP203v]	49.50%	0.00%	0.00%
	[SP204c, SP204v]	50.17%	3.24%	2.07%

- It was also shown that it is better to include as many synonyms as possible rather than using only one PPE word in the <Object> part of the compliance prompt.
- In contrast, in the case of violation prompts, good results were obtained when the prompts were minimally composed without using the term “hardhat” or synonyms and when negative sentences were not used.

Based on these findings, this study recommends composing a pair of prompts as follows:

- Compliance prompt: A description of a person using a term describing a PPE item and as many synonyms as possible.
- Violation prompt: A description of a person not using PPE-related terms in a positive sentence.

4.3. Performance comparison with few-shot object detection

Previous studies on construction site monitoring have used CNN models or variants to perform object detection tasks [2,4,8,57]. Limited research has been conducted on zero-shot/few-shot learning or standard datasets for PPE safety monitoring, making it difficult to directly compare the performance of the proposed method to previous studies. Therefore, this study devised an experiment to compare the proposed zero-shot method with general object detection method trained with few-shot image. It is for showing intuitive comparison on the performance between proposed method and traditional approaches. This study used YOLOv5 and YOLOv8 as comparison models for object detection because of their outstanding performance and wide adoption although those models are actually not designed for few-shot learning.

The YOLO models were trained repeatedly using 10, 20, 30, and 50 labeled images, and the performance levels were compared in terms of accuracy, precision, recall, F1-score, and F2-score metrics. For hardhat detection, only the labels of the hardhat and head were trained, and the training conditions were set to an image size of 640-pixel $\times$ 480-pixel, batch size of 5, and maximum training epoch of 500.

The results are represented in Table 7. The proposed zero-shot hardhat monitoring method outperformed the YOLOv5 and YOLOv8 models trained with 50 images, yielding average F2-scores of 85.6%, 74.4%, and 70.6%, respectively. Although it is difficult to compare directly due to the difference of the domain, the performance can be regarded as an improvement compared to the previous ZSL methods for construction site management, which demonstrated an average accuracy of 53% for four unseen classes [40]. Between YOLOv5 and YOLOv8, YOLOv8 exhibited a slightly lower performance than YOLOv5, which signifies recent models do not always perform better than old ones.

The performance of the YOLO models improved as more images were utilized in the training process. Therefore, it may be possible to improve hardhat detection performance by increasing the number of training images. However, since the purpose of this study is to find a way to eliminate the need for generating large datasets, especially for new PPE types, it is not in line with the aim of this study to increase the training data size as large as possible, solely focusing on performance

improvement.

4.4. Processing time for PPE detection

The time it takes to detect PPE was also measured. The proposed method has three steps compared to the traditional one-step object detection approach. Previous studies that monitored PPE using Faster R-CNN took between 0.206 s [2] and 0.362 s [4] per image. In contrast, studies using YOLOv5x, which is well known as its efficient processing time, only took 0.0165 s to process a single image [3].

Since the previous studies were carried out in diverse environments, we conducted a new experiment to gauge and compare the processing time of both the conventional and the proposed methods. We also used YOLOv5 and YOLOv8 models as the baseline models like the previous approach [3] on purpose of giving the reference of the experiment environment. All the tests were performed on a workstation equipped with an NVIDIA RTX 3080 10 GB GPU using 640-pixel by 640-pixel images to ensure consistency. The other conditions were the same as the other experiments.

Table 8 shows the mean processing time of the baseline models and the proposed method. As shown in Table 8, the proposed method took significantly longer time (1.7180 s) than YOLO models (0.0543 s and 0.0517 s for the YOLOv5 and YOLOv8 models, respectively). Nevertheless, it is important to note that the proposed PPE monitoring method does not aim to be used for an instant warning system that demands an immediate response. Instead, it aims to detect workers who violate PPE regulations. Considering this, an average of 1.7180 s for the proposed method is fast enough to achieve the objective.

4.5. Expanded application testing on other PPE types— safety goggles, vests, and gloves

Given that target objects can be detected from images without a training dataset, the same method can be utilized to recognize other objects. In this part of the study, in addition to the detection of hardhat wearing, the detection of safety goggles on the face, safety vests on the torso, and safety gloves on the hands were also tested. Before the experiment, the prompts for the PPE items were designated as follows:

“head, hardhat”: [“a person wearing a hardhat, protective headgear, or a helmet,” “a person with hair”].

Table 8
Average inference time comparison for PPE detection.

Method	Task	Mean processing time (s)
Baseline models	Object detection (YOLOv5)	0.0543
	Object detection (YOLOv8)	0.0517
	Human detection & body-part extraction	0.6722
Proposed method	Image-captioning	0.9884
	Test-similarity check	0.0594
	Total processing time	1.7180

Table 7

Performance comparison between the proposed zero-shot learning method and YOLOv5-based transfer learning method with a small dataset.

Method	Images	Accuracy	Precision	Recall	F1-score	F2-score
YOLOv5	10	49.73%	49.80%	72.00%	58.90%	66.11%
	20	59.20%	57.60%	69.30%	62.90%	66.59%
	30	63.56%	61.20%	74.00%	67.00%	71.03%
	50	73.08%	72.10%	75.00%	73.50%	74.40%
YOLOv8	10	50.00%	69.85%	32.57%	44.42%	36.46%
	20	51.27%	65.49%	54.12%	59.26%	56.07%
	30	59.03%	75.97%	60.29%	67.23%	62.89%
	50	72.26%	75.94%	69.42%	72.53%	70.63%
Proposed zero-shot monitoring	0	81.50%	78.04%	87.67%	82.57%	85.56%

“face, safety goggle”: [“a person wearing a goggle, or safety glasses,” “a person with a smile”].

“torso, safety vest”: [“a person wearing a vest, or a safety gear,” “a person wearing a shirt”].

“hand, safety glove”: [“a person wearing a safety glove,” “a person holding a tool”].

The prompt pairs were constructed according to the earlier method: a full sentence about wearing PPE, with synonyms, and a description of a person without PPE in a positive sentence. The prompt indicating the head part for hardhat detection was reused, and compliance prompts were written with the inclusion of synonyms for the target PPE following the clause “a person wearing.” Subsequently, violation prompts were written as positive sentences to describe PPE non-wearers. Because there is thus far no available dataset for the other PPEs (in contrast to the availability of a benchmark dataset for hardhats), we constructed a small dataset with 100 labeled for this experiment. The dataset included labels for safety goggles, vests, and gloves.

Table 9 outlines the performance results of the proposed PPE detection method on the other body parts. The F1 scores for safety goggles, vest, and glove detection were 69.23%, 76.80%, and 55.07%, respectively. Although these results were not as high as those of hardhat detection, they are still meaningful because they show that the proposed method can detect compliance with various PPE items without any training data, except for safety gloves.

Overall, our proposed zero-shot monitoring method demonstrated an F1-score of 82.57% for detecting non-hardhat wearers and 70.93% for overall workers’ PPE, including hardhats, goggles, vests, and gloves. Regarding the performance of other PPE, such as safety vests, goggles, and gloves, for which virtually no open datasets are available, our proposed method achieved an average F1-score of 67.03%. It is still meaningful performance of proposed ZSL considering YOLOv8 trained with 10 shots for those PPE resulted F1-score of 59.71%, 57.44%, and 54.30%, respectively.

5. Discussion

The proposed zero-shot-based method has strengths and limitations. In terms of strengths, it can achieve relatively very high performance without having large training datasets for various PPE types, compared to other zero-shot or few-shot methods (Section 4.3). Another strength is that it can be easily applied to other tasks, such as construction activity monitoring or even tasks outside of the construction domain, because it deploys general object detection, image captioning, and text similarity analysis as the three main modules. Similarly, the proposed method can be applied to monitoring of multiple workers, objects, and equipment items in a photo because it analyzes each object in a photo individually, according to the purpose of analysis.

Despite all these strengths, the proposed method also has some weaknesses. First, when a large dataset is available, the traditional object detection method performs better than the proposed method. For

example, some previous studies that deployed a large dataset for hardhat detection achieved an accuracy of approximately 95% [2–4]. Notwithstanding, traditional methods can only handle targeted PPE types with a large number of annotated datasets such as hardhats while the proposed method can any PPE types without a dataset.

Second, the proposed approach did not exhibit strong performance across all PPE types. As shown in Fig. 5, the proposed method can determine whether a worker complies with PPE regulations when automatically generated captions include terms directly or closely related to PPE items. For example, for hardhat monitoring, terms such as “helmet” and “hat,” which have a high cosine similarity with “hardhat,” are also allowable. For goggles monitoring, “glasses” are allowable, and for safety vest monitoring, even “apron” is allowable. However, in the case of gloves monitoring, gloves were often ignored in automatically generated captions, as listed in Fig. 5. This may be because gloves occupy a very small space in a photo and because bare hands and gloves can be difficult to distinguish, having similar colors and shapes.

Third, relying heavily on large pre-trained models, which can be computationally expensive and time-consuming to train, is another limitation of the method. As the performance of models used in this study improves, naturally the performance of the proposed method will likewise improve. However, one temporary solution to overcome this issue is to fine-tune the models used in this study using relatively small datasets rather than using a pure zero-shot approach.

6. Conclusion

Several studies have utilized CNN architectures for feature extraction in site monitoring [1–4,58]. However, the CNN approach requires a large dataset of labeled images [59], the construction of which is time-consuming and labor-intensive. To address this problem, this study proposes a zero-shot PPE monitoring model that does not require dataset preparation or model training. The proposed zero-shot method comprises the following three steps. First, it detects and extracts human-body key points using mm-Pose. It then crops human-body-part images based on the extracted body key points and generates captions for each body-part image using ClipCap. Semantic similarity scores between automatically generated image captions and predefined prompts based on the OSHA regulations were calculated using SBERT with MPNet as the backbone to determine worker compliance with the OSHA PPE regulations.

We conducted a series of experiments to validate our proposed method. For hardhat monitoring, which has been studied in several previous studies, the proposed zero-shot monitoring method (an F1-score of 82.57%) outperformed the few-shot object detection method using YOLO trained on 50 images (an F1-score of 62.9%). For monitoring of goggles, vests, and gloves, which have not been studied before and therefore do not have publicly available standard datasets, the performance of the proposed zero-shot method decreased to 76.8%, 69.23%, and 55.07%, respectively. Nevertheless, considering that these results were achieved without using any training data and also small gloves and bare hands in a picture are difficult to distinguish even by humans, the results are promising.

A major contribution of the proposed zero-shot method is that it significantly improves the zero-shot PPE monitoring, compared to both the previous ZSL method (an average accuracy of 53% for four unseen classes) [40] and the above-mentioned few-shot object detection models trained on 50 images. Another contribution of the proposed method is its generalizability and its ability to be expanded. As demonstrated in the expanded application tests, it can be applied to monitor compliance for hardhats and other PPE types, which do not have large annotated datasets. Moreover, it can be applied to monitor general objects and activities in domains beyond construction without any additional training or dataset, simply by changing the prompts. The third contribution is that the proposed method has a strength in that it can monitor multiple workers’ PPE compliance because its first step is to identify

Table 9

Performance of proposed zero-shot monitoring in checking compliance with various PPE items.

Target PPE item	Accuracy	Precision	Recall	F1-score	F2-score
Hardhat	81.50%	78.04%	87.67%	82.57%	85.56%
Safety goggle	61.90%	81.82%	60.00%	69.23%	63.38%
Safety vest	65.48%	78.69%	75.00%	76.80%	75.71%
Safety glove	48.33%	67.86%	46.34%	55.07%	49.48%
Mean (including hardhat)	64.60%	76.59%	67.26%	70.93%	68.54%
Mean (excluding hardhat)	58.57%	76.12%	60.45%	67.03%	62.86%

Input prompts

Head: ['a person wearing a hard hat, head protection, protective gear, or a helmet', 'a person with hair']

Face: ['a person wearing a goggle, or safety glasses', 'a person wearing a hat, or with hair']

Chest: ['a person wearing a vest, or a safety gear', 'a person wearing a shirt']

Hand: ['a person wearing a safety glove', 'a person holding a tool']

Input image	Automatically generated caption		Cosine similarity		Non-PPE wearer	
			Compliance.	Violation.	Actual	Pred.
	Hardhat	A man wearing a helmet and a red chain around his neck.	0.6551	0.3195	False	False
	Goggle	A man wearing glasses and a helmet with a red flower in his mouth.	0.5231	0.3830	False	False
	Safety Vest	A man in a yellow safety vest is holding a yellow object.	0.6522	0.3681	False	False
	Glove	A close up of a yellow and red Frisbee.	0.2221	0.1795	False	False
	Hardhat	A man wearing a yellow hat and a striped tie.	0.4585	0.4255	False	False
	Goggle	A man wearing a hat and a tie.	0.2929	0.6744	True	True
	Safety Vest	A man wearing a helmet and apron working on a construction site.	0.6155	0.3237	False	False
	Glove	A close up of a person holding a green apple.	0.3052	0.4318	False	True
	Hardhat	A man wearing a hat and sunglasses standing next to a traffic cone.	0.4459	0.2672	False	False
	Goggle	A close up of a person's face with a camera.	0.4037	0.4094	True	True
	Safety Vest	A man wearing a yellow vest and a black hat.	0.6044	0.3845	False	False
	Glove	A blurry photo of a person holding a cell phone.	0.4029	0.4021	True	False

Fig. 5. Comparison of the hardhat, goggle, safety vest, and glove cases.

each worker in a photo. This is then applied individually to each worker for PPE monitoring.

The critical limitation of the proposed method is that its performance relies significantly on the applied large pre-trained models and how the large pre-trained model was trained based on these datasets. Nevertheless, as the performance of large pre-trained models improves exponentially every year, the performance of the proposed image-captioning-based zero-shot monitoring method will also increase accordingly. Currently, there is ample room for improvement. For instance, prompt engineering can further enhance performance [42], this study did not focus on prompt engineering, although. In addition, it is necessary to consider external environmental factors such as weather and lighting conditions, which affect the object detection performance [2,4]. These need to be addressed in future research.

In conclusion, the proposed zero-shot monitoring method is a significant improvement over previous methods and has the potential to be applied to other domains. Although there are limitations, such as reliance on the performance of the applied large pre-trained model, further development can improve its stability and practicality in the field.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korean government (Ministry of

Science and ICT, MSIT) (No. 2021R1A2C3008209).

References

- [1] V.S.K. Delhi, R. Sankarlal, A. Thomas, Detection of personal protective equipment (PPE) compliance on construction site using computer vision based deep learning techniques, *Front. Built Environ.* 6 (2020), <https://doi.org/10.3389/fbuilt.2020.00136>.
- [2] Q. Fang, H. Li, X. Luo, L. Ding, H. Luo, T.M. Rose, W. An, Detecting non-hardhat-use by a deep learning method from far-field surveillance videos, *Autom. Constr.* 85 (2018) 1–9, <https://doi.org/10.1016/j.autcon.2017.09.018>.
- [3] A. Hayat, F. Morgado-Dias, Deep learning-based automatic safety helmet Detection system for construction safety, *Appl. Sci.* 12 (2022) 8268, <https://doi.org/10.3390/app12168268>.
- [4] J. Shen, X. Xiong, Y. Li, W. He, P. Li, X. Zheng, Detecting safety helmet wearing on construction sites with bounding-box regression and deep transfer learning, *Comput.-Aided Civ. Infrastruct. Eng.* 36 (2021) 180–196, <https://doi.org/10.1111/mice.12579>.
- [5] H. Kim, H. Kim, Y.W. Hong, H. Byun, Detecting construction equipment using a region-based fully convolutional network and transfer learning, *J. Comput. Civ. Eng.* 32 (2018) 04017082, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000731](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000731).
- [6] B. Xiao, S.-C. Kang, Development of an image data set of construction Machines for Deep Learning Object Detection, *J. Comput. Civ. Eng.* 35 (2021) 05020005, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000945](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000945).
- [7] H. Tajeen, Z. Zhu, Image dataset development for measuring construction equipment recognition performance, *Autom. Constr.* 48 (2014) 1–10, <https://doi.org/10.1016/j.autcon.2014.07.006>.
- [8] J. Wu, N. Cai, W. Chen, H. Wang, G. Wang, Automatic detection of hardhats worn by construction personnel: a deep learning approach and benchmark dataset, *Autom. Constr.* 106 (2019) 102894, <https://doi.org/10.1016/j.autcon.2019.102894>.
- [9] N.D. Nath, T. Chaspari, A.H. Behzadan, Single- and multi-label classification of construction objects using deep transfer learning methods, *J. Inf. Technol. Constr.* 24 (2019) 511–526, <https://doi.org/10.36680/j.itcon.2019.028>.
- [10] L. Chen, Y. Wang, M.-F.F. Siu, Detecting semantic regions of construction site images by transfer learning and saliency computation, *Autom. Constr.* (2020) 103185, <https://doi.org/10.1016/j.autcon.2020.103185>.
- [11] S. Bang, F. Baek, S. Park, W. Kim, H. Kim, Image augmentation to improve construction resource detection using generative adversarial networks, cut-and-paste, and image transformation techniques, *Autom. Constr.* 115 (2020) 103198, <https://doi.org/10.1016/j.autcon.2020.103198>.

- [12] M. Alawadhi, W. Yan, B.I.M. Hyperreality, Data synthesis using BIM and Hyperrealistic rendering for deep, Learning (2021), <https://doi.org/10.48550/arXiv.2105.04103>.
- [13] F. Baek, S. Park, H. Kim, Data Augmentation Using Adversarial Training for Construction-Equipment Classification, 2019, <https://doi.org/10.48550/arXiv.1911.11916>.
- [14] J. Kim, S. Chi, A few-shot learning approach for database-free vision-based monitoring on construction sites, *Autom. Constr.* 124 (2021) 103566, <https://doi.org/10.1016/j.autcon.2021.103566>.
- [15] M. Kampffmeyer, Y. Chen, X. Liang, H. Wang, Y. Zhang, E.P. Xing, Rethinking knowledge graph propagation for zero-shot learning, in: *ArXiv180511724 Cs*, 2019, <https://doi.org/10.48550/arXiv.1805.11724>. Long Beach, California, USA.
- [16] A. Radford, J.W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, I. Sutskever, Learning transferable visual models from natural language supervision, in: *Proc. 38th Int. Conf. Mach. Learn.*, PMLR, 2021, pp. 8748–8763, in: <https://proceedings.mlr.press/v139/radford21a.html>.
- [17] B.L. Suderman, R.W. Hoover, R.P. Ching, I.S. Scher, The effect of hardhats on head and neck response to vertical impacts from large construction objects, *Accid. Anal. Prev.* 73 (2014) 116–124, <https://doi.org/10.1016/j.aap.2014.08.011>.
- [18] M.-W. Park, N. Elsafty, Z. Zhu, Hardhat-wearing Detection for enhancing on-site safety of construction workers, *J. Constr. Eng. Manag.* 141 (2015) 04015024, [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0000974](https://doi.org/10.1061/(ASCE)CO.1943-7862.0000974).
- [19] Z. Zhu, M.-W. Park, N. Elsafty, Automated Monitoring of Hardhats Wearing for Onsite Safety Enhancement, 2015, <https://doi.org/10.14288/1.0076342>.
- [20] S. Du, M. Shehata, W. Badawy, Hard hat detection in video sequences based on face features, motion and color information, in: *2011 3rd Int. Conf. Comput. Res. Dev.*, Shanghai, China, 2011, pp. 25–29, <https://doi.org/10.1109/ICCRD.2011.5763846>.
- [21] A.H.M. Rubaiyat, T.T. Toma, M. Kalantari-Khandani, S.A. Rahman, L. Chen, Y. Ye, C.S. Pan, Automatic detection of helmet uses for construction safety, in: *2016 IEEE WICACM Int. Conf. Web Intell. Workshop WIW*, 2016, pp. 135–142, <https://doi.org/10.1109/WIW.2016.045>.
- [22] M.-W. Park, I. Brilakis, Construction worker detection in video frames for initializing vision trackers, *Autom. Constr.* 28 (2012) 15–25, <https://doi.org/10.1016/j.autcon.2012.06.001>.
- [23] S. Ren, K. He, R. Girshick, J. Sun, Faster R-CNN: towards real-time object Detection with region proposal, *Networks* (2016), <https://doi.org/10.48550/arXiv.1506.01497>.
- [24] J. Redmon, S. Divvala, R. Girshick, A. Farhadi, You Only Look Once: unified, real-time object detection, in: *2016 IEEE Conf. Comput. Vis. Pattern Recognit. CVPR*, Las Vegas, Nevada, USA, 2016, pp. 779–788, https://www.cv-foundation.org/openaccess/content_cvpr_2016/html/Redmon_You_Only_Look_CVPR_2016_paper.html.
- [25] G. Huang, Z. Liu, L.V.D. Maaten, K.Q. Weinberger, Densely connected convolutional networks, in: *2017 IEEE Conf. Comput. Vis. Pattern Recognit. CVPR*, IEEE Computer Society, Honolulu, Hawaii, USA, 2017, pp. 2261–2269, <https://doi.org/10.1109/CVPR.2017.243>.
- [26] Z. Kolar, H. Chen, X. Luo, Transfer learning and deep convolutional neural networks for safety guardrail detection in 2D images, *Autom. Constr.* 89 (2018) 58–70, <https://doi.org/10.1016/j.autcon.2018.01.003>.
- [27] Z. Zheng, Z. Zhang, W. Pan, Virtual prototyping- and transfer learning-enabled module detection for modular integrated construction, *Autom. Constr.* 120 (2020) 103387, <https://doi.org/10.1016/j.autcon.2020.103387>.
- [28] S. Bunrit, N. Kerdprasop, K. Kerdprasop, Evaluating on the transfer learning of CNN architectures to a construction material image classification task, *Int. J. Mach. Learn. Comput.* 9 (2019) 201–207, <https://doi.org/10.18178/ijmlc.2019.9.2.787>.
- [29] J. Zhu, C. Zhang, H. Qi, Z. Lu, Vision-based defects detection for bridges using transfer learning and convolutional neural networks, *Struct. Infrastruct. Eng.* 16 (2020) 1037–1049, <https://doi.org/10.1080/15732479.2019.1680709>.
- [30] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, L. Fei-Fei, ImageNet: A large-scale hierarchical image database, in: *2009 IEEE Conf. Comput. Vis. Pattern Recognit*, 2009, pp. 248–255, <https://doi.org/10.1109/CVPR.2009.5206848>.
- [31] T.-Y. Lin, M. Maire, S. Belongie, L. Bourdev, R. Girshick, J. Hays, P. Perona, D. Ramanan, C.L. Zitnick, P. Dollár, Microsoft COCO: Common Objects in Context, *ArXiv14050312 Cs* (2015), <https://doi.org/10.48550/arXiv.1405.0312>.
- [32] R. Duan, H. Deng, M. Tian, Y. Deng, J. Lin, SODA: a large-scale open site object detection dataset for deep learning in construction, *Autom. Constr.* 142 (2022) 104499, <https://doi.org/10.1016/j.autcon.2022.104499>.
- [33] N.M. Müller, K. Markert, Identifying Mislabelled Instances in Classification Datasets, in: *2019 Int. Jt. Conf. Neural Netw. IJCNN*, Budapest, Hungary, 2019, pp. 1–8, <https://doi.org/10.1109/IJCNN.2019.8851920>.
- [34] G. Koch, R. Zemel, R. Salakhutdinov, Siamese neural networks for one-shot image recognition, in: *ICML Deep Learn. Workshop*, Lille, 2015, <https://www.cs.cmu.edu/~rsalakhu/papers/oneshot1.pdf>.
- [35] J. Snell, K. Swersky, R. Zemel, Prototypical Networks for few-shot learning, in: *Adv. Neural Inf. Process. Syst.*, Curran Associates, Inc., 2017, in: <https://proceedings.neurips.cc/paper/2017/hash/cb8da6767461f2812ae4290eac7cbc42-Abstract.html>.
- [36] C. Finn, P. Abbeel, S. Levine, Model-agnostic meta-learning for fast adaptation of deep networks, in: *Proc. 34th Int. Conf. Mach. Learn.*, PMLR, 2017, pp. 1126–1135, in: <https://proceedings.mlr.press/v70/finn17a.html>.
- [37] Z. Cui, Q. Wang, J. Guo, N. Lu, Few-shot classification of façade defects based on extensible classifier and contrastive learning, *Autom. Constr.* 141 (2022) 104381, <https://doi.org/10.1016/j.autcon.2022.104381>.
- [38] B. Romera-Paredes, P. Torr, An embarrassingly simple approach to zero-shot learning, in: *Proc. 32nd Int. Conf. Mach. Learn.*, PMLR, 2015, pp. 2152–2161, in: <https://proceedings.mlr.press/v37/romera-paredes15.html>.
- [39] F. Pourpanah, M. Abdar, Y. Luo, X. Zhou, R. Wang, C.P. Lim, X.-Z. Wang, Q.M. J. Wu, A review of generalized zero-shot learning methods, *IEEE Trans. Pattern Anal. Mach. Intell.* (2022) 4051–4070, <https://doi.org/10.1109/TPAMI.2022.3191696>.
- [40] Z. Pan, C. Su, Y. Deng, J. Cheng, Video2Entities: a computer vision-based entity extraction framework for updating the architecture, engineering and construction industry knowledge graphs, *Autom. Constr.* 125 (2021) 103617, <https://doi.org/10.1016/j.autcon.2021.103617>.
- [41] U. Mall, B. Hariharan, K. Bala, Zero-shot Learning Using Multimodal Descriptions, in: *2022 IEEE CVF Conf. Comput. Vis. Pattern Recognit, Workshop CVPRW*, IEEE, New Orleans, LA, USA, 2022, pp. 3930–3938, <https://doi.org/10.1109/CVPRW56347.2022.00438>.
- [42] X.L. Li, P. Liang, Prefix-Tuning: Optimizing Continuous Prompts for Generation, 2021, <https://doi.org/10.48550/arXiv.2101.00190>.
- [43] G. Yong, K. Jeon, D. Gil, G. Lee, Prompt engineering for zero-shot and few-shot defect detection and classification using a visual-language pretrained model, *Comput.-Aided Civ. Infrastruct. Eng.* (2022), <https://doi.org/10.1111/mice.12954>.
- [44] Personal Protective Equipment - Standards | Occupational Safety and Health Administration. <https://www.osha.gov/personal-protective-equipment/standards>, 2024.
- [45] A. Sengupta, F. Jin, R. Zhang, S. Cao, Mm-pose: real-time human skeletal posture estimation using mmWave radars and CNNs, *IEEE Sensors J.* 20 (2020) 10032–10044, <https://doi.org/10.1109/JSEN.2020.2991741>.
- [46] J. Wang, K. Sun, T. Cheng, B. Jiang, C. Deng, Y. Zhao, D. Liu, Y. Mu, M. Tan, X. Wang, W. Liu, B. Xiao, Deep High-Resolution Representation Learning for Visual Recognition, 2020, <https://doi.org/10.48550/arXiv.1908.07919>.
- [47] R. Mokady, A. Hertz, A.H. Bermano, ClipCap: CLIP Prefix for Image Captioning, 2021, <https://doi.org/10.48550/arXiv.2111.09734>.
- [48] P. Sharma, N. Ding, S. Goodman, R. Soiccut, Conceptual captions: A cleaned, Hypernymed, image alt-text dataset for automatic image captioning, in: *Proc. 56th Annu. Meet. Assoc. Comput. Linguist. vol. 1*, Long Pap., Association for Computational Linguistics, Melbourne, Australia, 2018, pp. 2556–2565, <https://doi.org/10.18653/v1/P18-1238>.
- [49] S. Jiang, J. Zhang, Real-time crack assessment using deep neural networks with wall-climbing unmanned aerial system, *Comput.-Aided Civ. Infrastruct. Eng.* 35 (2020) 549–564, <https://doi.org/10.1111/mice.12519>.
- [50] T. Schick, H. Schütze, Exploiting cloze questions for few shot text classification and natural language inference, *ArXiv200107676 Cs* (2021), <https://doi.org/10.48550/arXiv.2001.07676>.
- [51] P. Liu, W. Yuan, J. Fu, Z. Jiang, H. Hayashi, G. Neubig, Pre-train, Prompt, and Predict: A Systematic Survey of Prompting Methods in Natural Language Processing, *ACM Comput. Surv.* 55 (195) (2023) 1–195, 35, <https://doi.org/10.1145/3560815>.
- [52] V. Liu, L.B. Chilton, Design guidelines for prompt engineering text-to-image generative models, in: *CHI Conf. Hum. Factors Comput. Syst.*, ACM, New Orleans LA USA, 2022, pp. 1–23, <https://doi.org/10.1145/3491102.3501825>.
- [53] Collins English Dictionary, in: *Collins Engl. Dict*, 2021. <https://www.collinsdictionary.com/dictionary/english/protective-headgear>.
- [54] N. Reimers, I. Gurevych, Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks, *ArXiv190810084 Cs* (2019), <https://doi.org/10.48550/arXiv.1908.10084>.
- [55] K. Song, X. Tan, T. Qin, J. Lu, T.-Y. Liu, MPNet: Masked and permuted pre-training for language understanding, in: *Proc. 34th Int. Conf. Neural Inf. Process. Syst.*, Curran Associates Inc., Red Hook, NY, USA, 2020, pp. 16857–16867.
- [56] L. Xie, Hardhat, Harvard Dataverse, 2019, <https://doi.org/10.7910/DVN/7CBGOS>.
- [57] B. Xiao, Q. Lin, Y. Chen, A vision-based method for automatic tracking of construction machines at nighttime based on deep learning illumination enhancement, *Autom. Constr.* 127 (2021) 103721, <https://doi.org/10.1016/j.autcon.2021.103721>.
- [58] J. Liu, H. Luo, L. Liu, Deep learning-based data analytics for safety in construction, *Autom. Constr.* 140 (2022) 104302, <https://doi.org/10.1016/j.autcon.2022.104302>.
- [59] A. Parnami, M. Lee, Learning from few examples: a summary of approaches to few-shot, *Learning* (2022), <https://doi.org/10.48550/arXiv.2203.04291>.